SKILL-6 : Spring MVC Web Request Handling Demo

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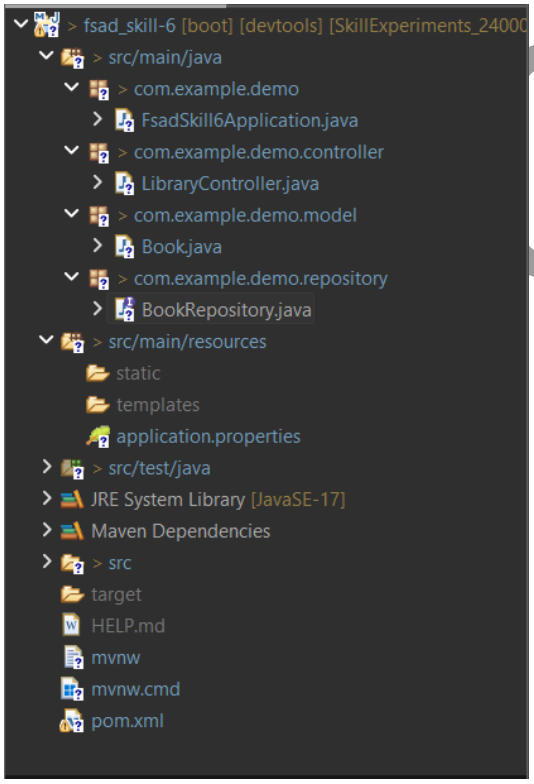
Section: 2

Scenario:

An online library system wants to allow users to explore books, view specific book details, submit feedback, and interact with the system through various types of requests. To support these features, the backend team needs to implement multiple Spring MVC endpoints demonstrating request mappings, path variables, request parameters, and JSON data handling. Your task is to create a Spring Boot MVC controller that exposes different demo operations using @RestController, @GetMapping, @PostMapping, and @PathVariable.

Structure:



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Codes:

Book.java:

package com.example.demo.model;

```
import jakarta.persistence.Entity;
import jakarta.persistence.GeneratedValue;
import jakarta.persistence.GenerationType;
import jakarta.persistence.Id;
import jakarta.persistence.Table;

@Entity
@Table(name = "book_skill6")
public class Book
{
  @Id
  @GeneratedValue(strategy = GenerationType.IDENTITY)
  private Integer id;

  private String title;
  private String author;
  private double price;

  public Book() {}

  public Book(Integer id, String title, String author, double price)
  {
    this.id = id;
    this.title = title;
    this.author = author;
    this.price = price;
  }

  public Integer getId()
  {
    return id;
```

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```
}

public void setId(Integer id)
{
   this.id = id;
}

public String getTitle()
{
   return title;
}

public void setTitle(String title)
{
   this.title = title;
}

public String getAuthor()
{
   return author;
}

public void setAuthor(String author)
{
   this.author = author;
}

public double getPrice()
{
   return price;
}
```

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```
    public void setPrice(double price)
    {
        this.price = price;
    }
}
```

LibraryController.java:

```
package com.example.demo.controller;

import java.util.Arrays;
import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.*;

import com.example.demo.model.Book;
import com.example.demo.repository.BookRepository;

@RestController
public class LibraryController
{

    @Autowired
    BookRepository repo;

    @GetMapping("/welcome")
    public String welcome()
    {
        return "Welcome to Online Library System";
    }
```

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```
@GetMapping("/count")
public long getBookCount()
{
    return repo.count();
}

@GetMapping("/price")
public double getPrice()
{
    return 499.99;
}

@GetMapping("/books")
public List<String> getBooks()
{
    return Arrays.asList("Java Programming", "Spring Boot Guide", "Data Structures", "Python Basics");
}

@GetMapping("/books/{id}")
public Book getBookById(@PathVariable int id)
{
    return repo.findById(id).orElse(null);
}

@GetMapping("/search")
public String searchBook(@RequestParam String title)
{
    return "Searching for book titled : " + title;
}
```

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```
    @GetMapping("/author/{name}")
    public String getAuthor(@PathVariable String name)
    {
        return "Books written by Author : " + name;
    }

    @PostMapping("/addbook")
    public Book addBook(@RequestBody Book book)
    {
        return repo.save(book);
    }

    @GetMapping("/viewbooks")
    public List<Book> viewBooks()
    {
        return repo.findAll();
    }

}
```

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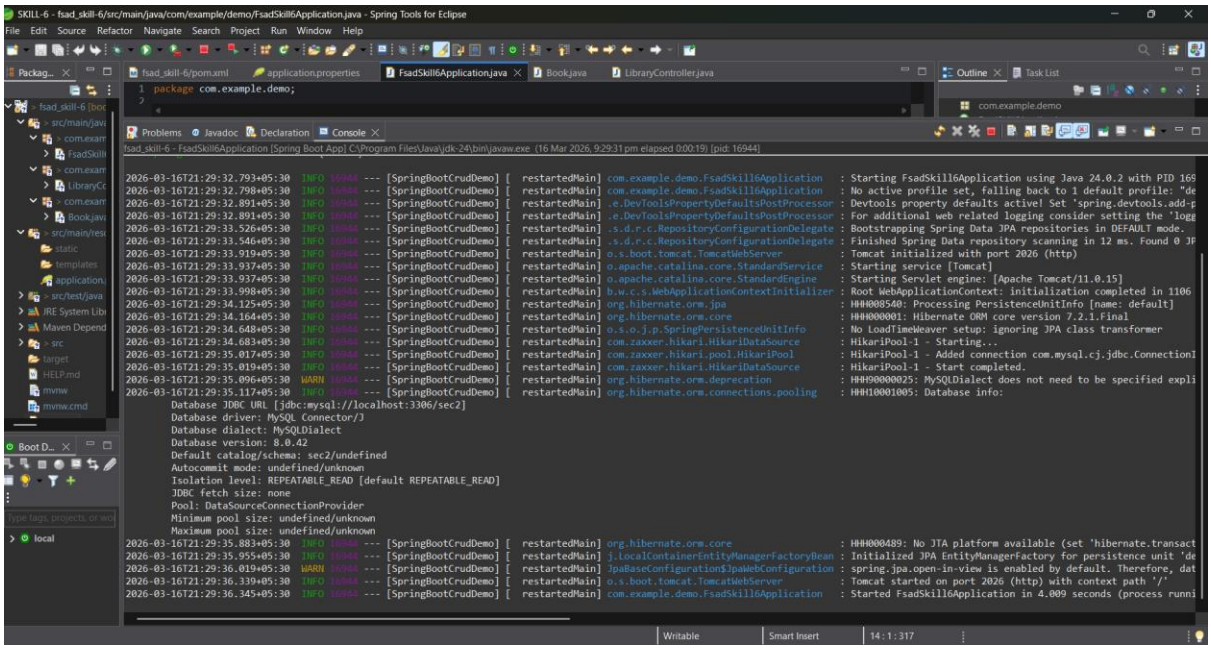
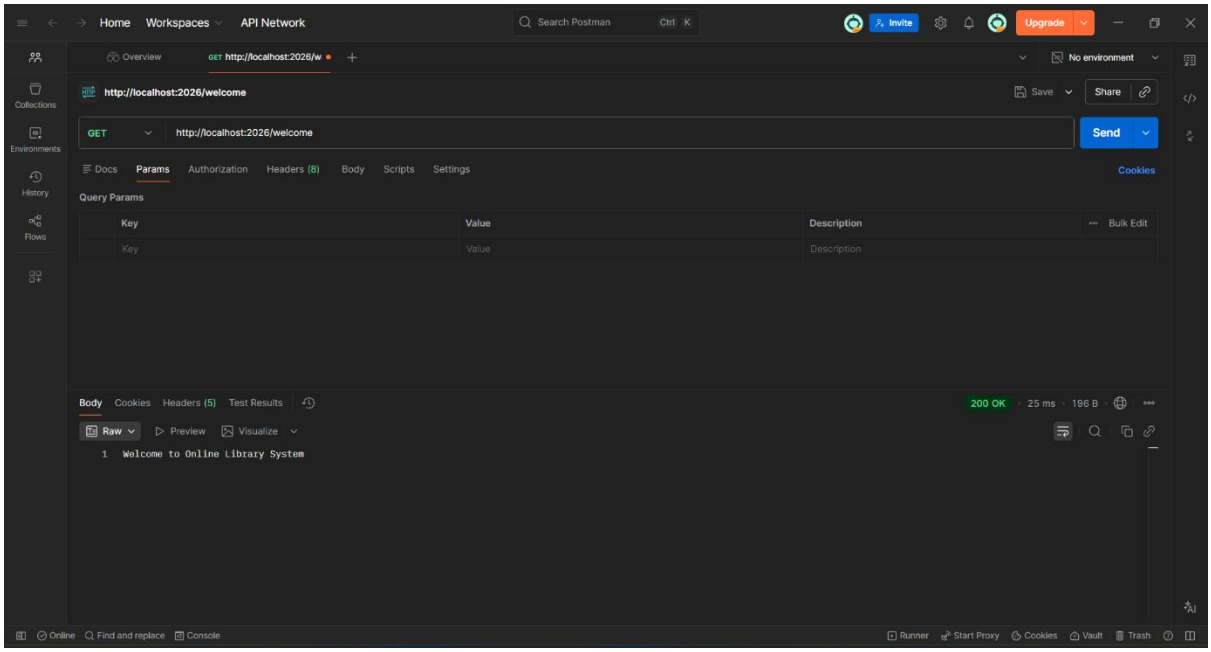
BookRepository.java:

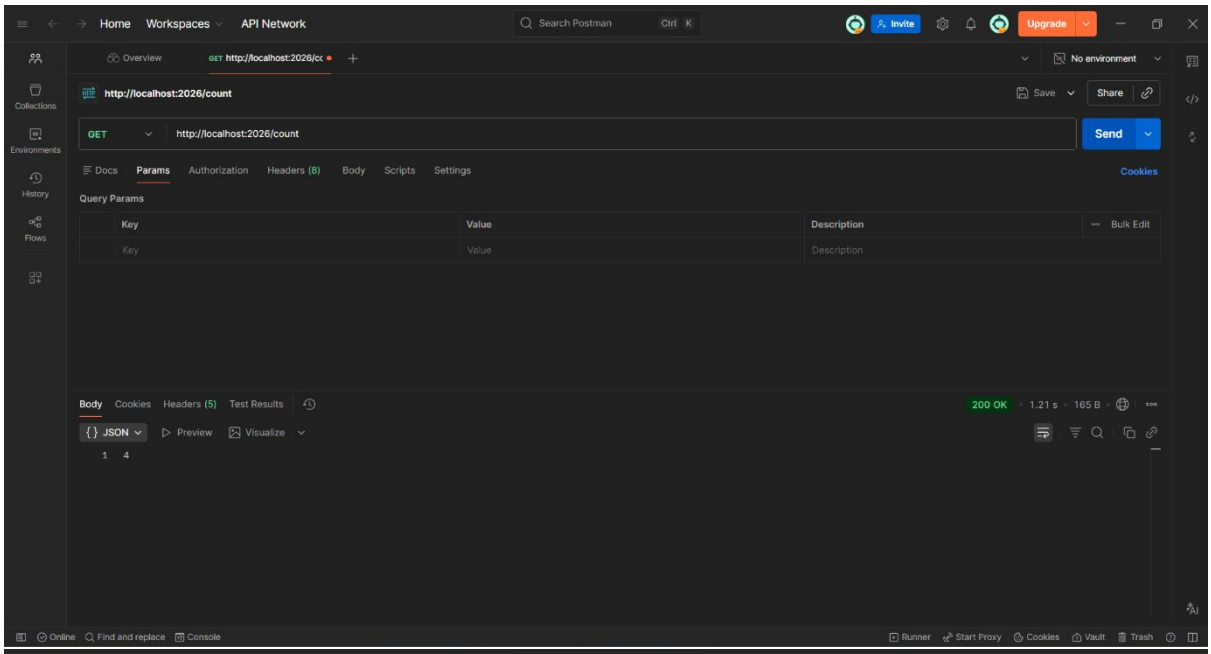
```
package com.example.demo.repository;

import org.springframework.data.jpa.repository.JpaRepository;
import com.example.demo.model.Book;

public interface BookRepository extends JpaRepository<Book, Integer>
{

}
```

OUTPUTS:



http://localhost:2026/welcome




Home
Workspaces
API Network
Search Postman
Invite
Upgrade
Overview
GET http://localhost:2026/cc
No environment
Collections
Environments
History
Flows
http://localhost:2026/count
Save
Share
GET
http://localhost:2026/count
Send
Docs
Params
Authorization
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Description
Bulk Edit
Body
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1 4
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Find and replace
Console
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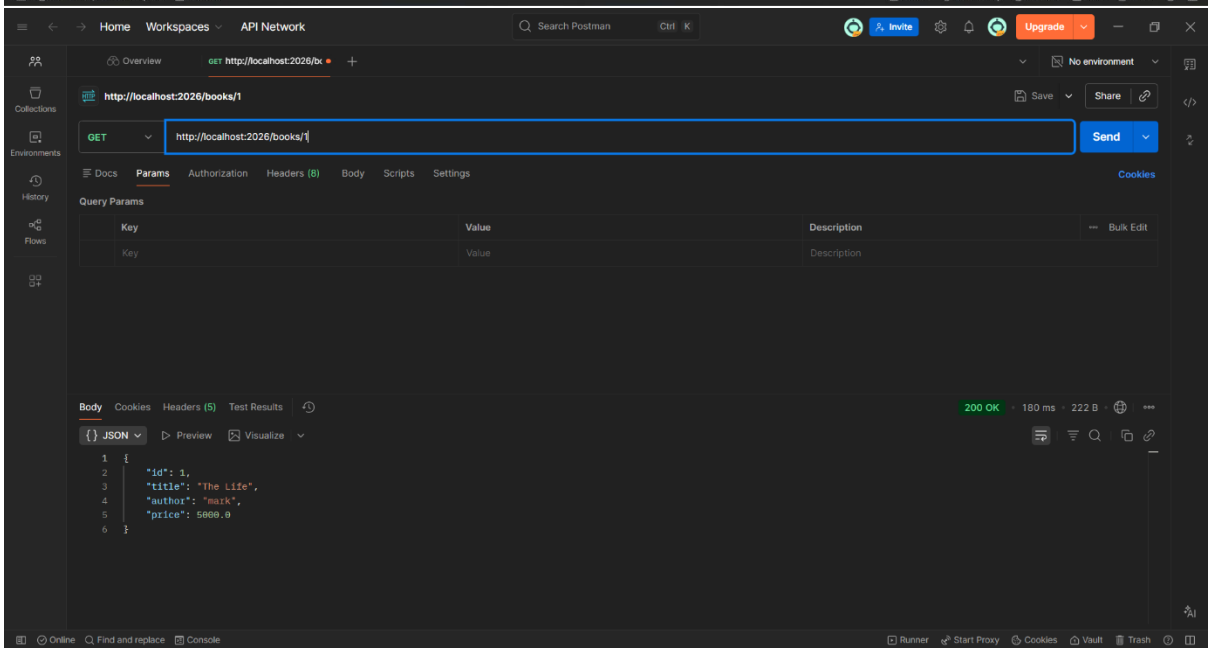
Start Proxy
Cookies
Vault
Trash


Home
Workspaces
API Network
Search Postman
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No environment
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http://localhost:2026/books/1
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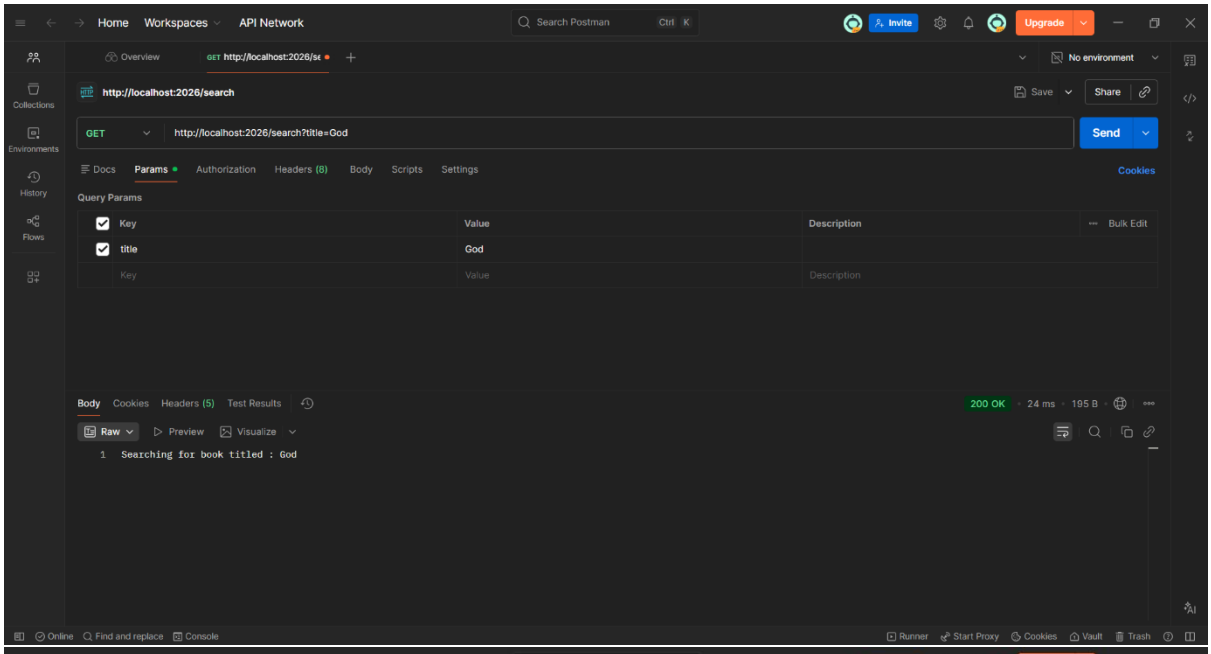
Headers (8)
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
Body
Cookies
Headers (5)
Test Results
200 OK
180 ms
222 B
JSON
Preview
Visualize
1 {
2 "id": 1,
3 "title": "The Life",
4 "author": "mark",
5 "price": 5000.0
6 }
Online
Find and replace
Console
Runner
Start Proxy
Cookies
Vault
Trash


Home
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No environment
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GET
http://localhost:2026/search?title=God
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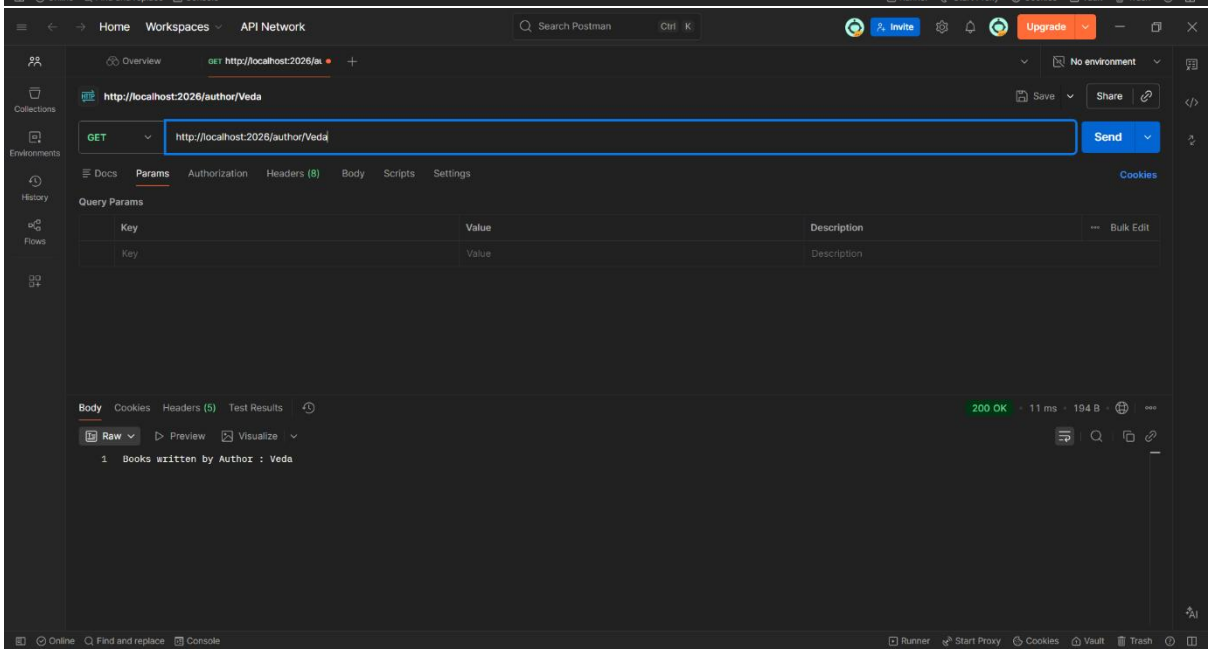
Headers (8)
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
title
God
Body
Cookies
Headers (5)
Test Results
200 OK
24 ms
195 B
Raw
Preview
Visualize
1 Searching for book titled : God
Online
Find and replace
Console
Runner
Start Proxy
Cookies
Vault
Trash


Home
Workspaces
API Network
Search Postman
Invite
Upgrade
Overview
GET http://localhost:2026/au
No environment
Collections
Environments
History
Flows
http://localhost:2026/author/Veda
Save
Share
GET
http://localhost:2026/author/Veda
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Authorization
Headers (8)
Body
Scripts
Settings
Cookies
Query Params
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Description
Bulk Edit
Body
Cookies
Headers (5)
Test Results
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Preview
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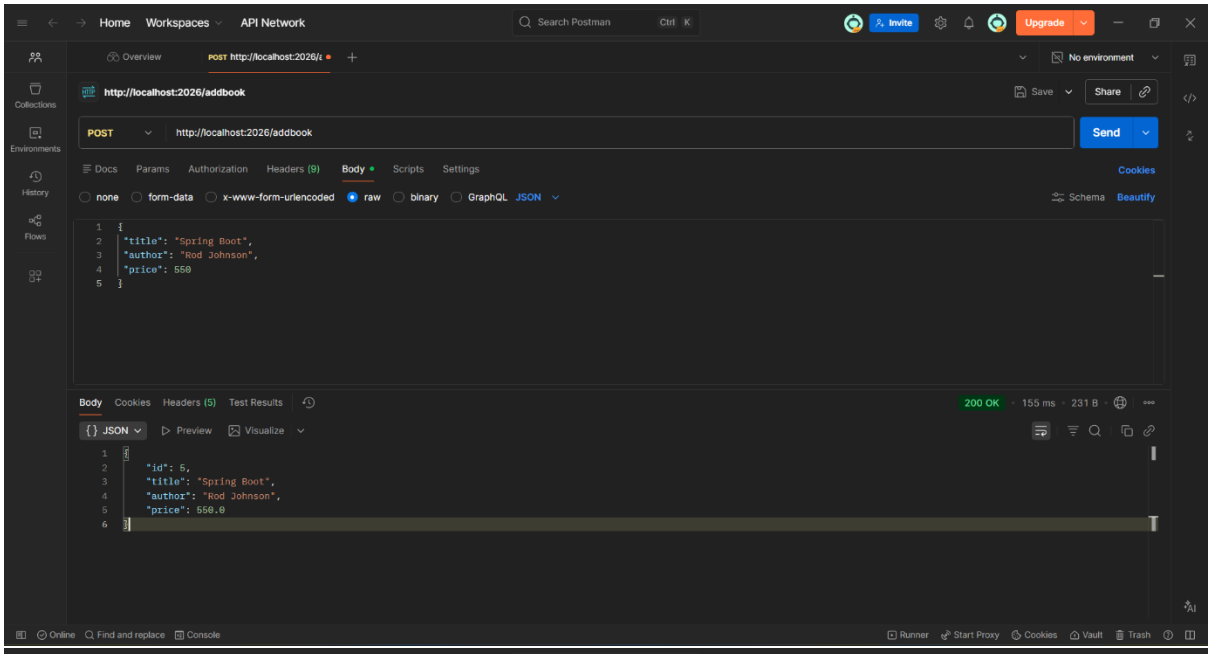
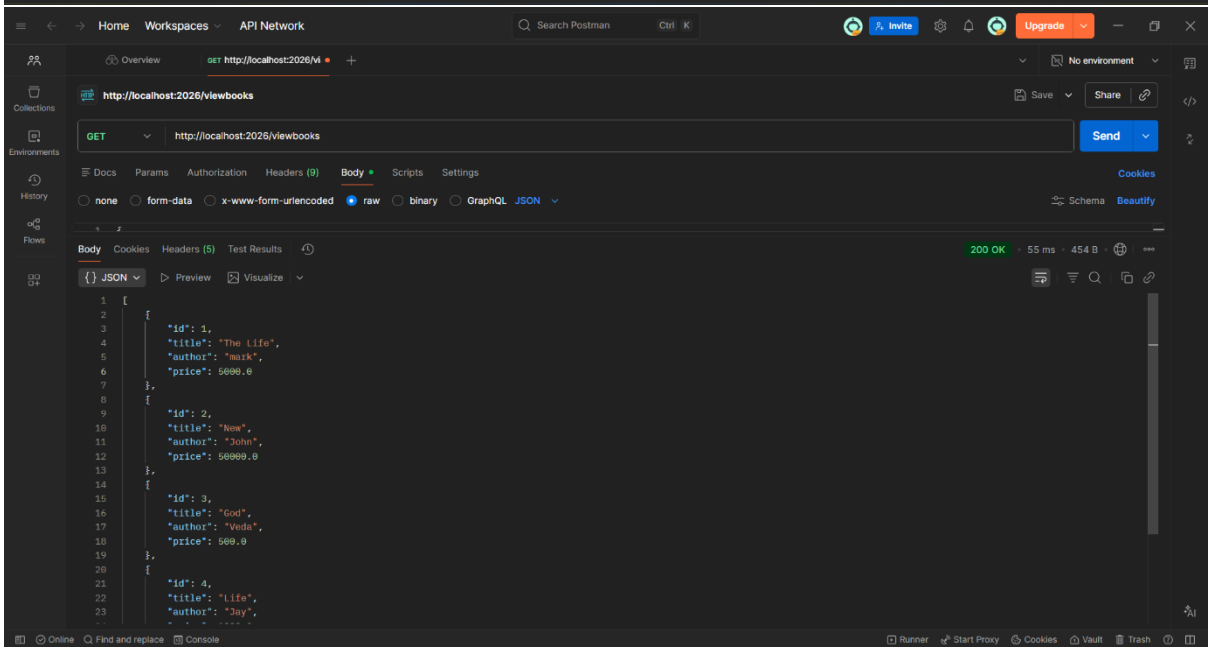
1 Books written by Author : Veda
Online
Find and replace
Console
Runner
Start Proxy
Cookies
Vault
Trash





GITHUB LINK:

https://github.com/nagasubrahmanyammeka/SkillExperiments_2400030128_FSAD_Sec2/tree/main/SKILL-6